

Dynamic Prompt-Driven Zero-Shot Relation Extraction

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Abstract—The task of zero-shot relation extraction is a very important research topic in the field of information extraction, which can effectively alleviate the issue of no training samples for some relations. Existing zero-shot relation extraction methods based on pre-trained language models (PLMs) always extract features from sentences, but cannot provide satisfactory semantic representations when there are conflicts between the task and the knowledge contained in PLMs. To address the issue, based on Prompt paradigm, a novel dynamic Prompt-driven method is proposed, aimed at fully stimulating the knowledge from PLMs to promote relation extraction. Specifically, the task of zero-shot relation extraction is defined as a masked language model (MLM) task, where [MASK] representation is qualified as relation representation for classification. Further, the key problems of Prompt paradigm for zero-shot relation extraction are explored, including the effect of template in Prompt and representation degradation. On this basis, in the new model, we utilize dynamic template to provide greater flexibility and introduce contrastive learning to optimize the semantic representation. Extensive experiments are conducted on three benchmark datasets (FewRel, TACRED, and Wiki-ZSL), demonstrating that the proposed model achieves the state-of-the-art performance with solving the existing problems.

Index Terms—Relation extraction, zero-shot learning, Prompt paradigm, dynamic template.

I. INTRODUCTION

THE task of relation extraction is an important basic task in the field of information extraction, aimed at recognizing the link type between entities in a certain context. The task facilitates the development of several downstream tasks, such as knowledge graph construction, intelligent answer system, machine translation, and reading comprehension [1], and thus has been widely concerned. For such a task, early methods [2], [3] often employ simple neural networks like convolutional neural network (CNN) and long-short term memory network (LSTM) to capture semantic distinction. Afterward, numerous innovative methods emerged, such as TPLinker [4] leveraging token-based link prediction and RIFRE [5] utilizing heterogeneous graph structure representation. However, the above methods always

require a large amount of labeled data. Due to the absence of training samples for most relation types in the real world, zero-shot relation extraction becomes the focus of attention.

Zero-shot relation extraction task is to learn a model based on labeled data with seen relations and further recognize unseen relations. Early researches always emphasize the strategy of problem transformation, in which the main idea is to convert this task into other problems. Specifically, question answering [6] and text entailment [7] are two prominent examples. However, these strategies are limited by the ability of text representation, until pre-trained language models (PLMs) is developed. Spearheaded by BERT [8], PLMs are learned by large-scale corpus and possess exceptional text representation capabilities. Sequentially, some methods [9] identify unseen relations by mapping contextual sentence and relation descriptions into the same semantic space. Despite the initial success, the knowledge contained in PLMs has not been effectively utilized. It is necessary to explore more effective ways for utilizing PLMs in the task of zero-shot relation extraction.

Currently, Prompt paradigm has become a popular tool for applying PLMs. It is a learning technology to stimulate the knowledge, which is hidden in PLMs and required by downstream tasks. Concretely, Prompt converts the original task into a new one, which is the same as the pre-trained task of PLMs. Such an approach is always able to perform well on specific tasks. Previous research [10] demonstrates that Prompt paradigm can improve the efficiency of sample data usage in low-resource situations, and thus Prompt has been introduced to few-shot relation extraction [11]. Lately, to alleviate the lack of training data for unseen relations, Prompt is beginning to be applied in zero-shot relation extraction. For instance, ZS-SKA [12] guides the mapping of sentence semantic and relation description in Prompt paradigm through knowledge graph. RelationPrompt [13] uses Prompt paradigm to directly extract relations by augmenting the data via text generation models. Both methods alleviate the problem of lack of training data for unseen relations in zero-shot tasks to some extent. However, these methods have to introduce noise during introducing additional data, and it is difficult to balance the noise and effective information. This issue should be avoided by delving deeper into the knowledge included inside PLMs.

In this paper, we begin by investigating the possible causes for low accuracy when employing Prompt paradigm to relation extraction. Based on an implemented model on Prompt-driven zero-shot relation extraction, we conduct some preliminary studies to reveal three key points: (1) the embedding generated by

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the pre-trained language model can be used directly as a relation representation, (2) the templates in Prompt have a greater impact on the representation of semantic space, and (3) Prompt-based masked language models suffer from representation degradation issue. In view of the above experiences and issues, we propose a **Dynamic Prompt-Driven Model (DPDM)**, which combines dynamic template in Prompt with prototype learning for zero-shot relation extraction. Specifically, zero-shot relation extraction is defined as a MLM task, where the [MASK] representation is qualified as a relation representation for classification. To apply Prompt more effectively, we firstly devise and adopt dynamic template for zero-shot relation extraction, which can provide greater flexibility in the case of low data resources. Besides, to optimize the semantic representation, a contrastive learning strategy on relation representation is introduced. Finally, a series of experiments is conducted on three benchmark datasets, to verify the effectiveness of the proposed model.

The main contributions of this work can be summarized as follows.

- 1) A series of preliminary studies is conducted that reveals the main challenges encountered by Prompt paradigm in zero-shot relation extraction.
- 2) A novel dynamic Prompt-driven model for zero-shot relation extraction is proposed. To our knowledge, dynamic template is firstly devised and introduced in such a task, which breaks through the bottleneck of manually-design templates.
- 3) Extensive experiments are conducted on three benchmark datasets, demonstrating that the proposed model achieves the state-of-the-art performance with strong robustness.

The structure of this paper is organized as follows. Section II summarizes the development of Prompt paradigm in the field of natural language processing and related work on zero-shot relation extraction. In section III, the preliminary studies are illustrated, to explore the key issues on applying Prompt for zero-shot relation extraction. Based on them, Section IV further elaborates the details of the proposed model. Extensive experiments are conducted to verify the effectiveness and robustness of the proposed model in Section V, before providing a brief conclusion in Section VI.

II. RELATED WORK

A. Prompt Learning in NLP

Prompt paradigm works based on pre-trained language models. With the development of pre-trained language models like BERT [8] and RoBERTa [14], the field of natural language processing (NLP) has reached a new era. Pre-trained language model is a type of model with a large number of parameters that have been trained by self-supervised tasks like masked language model (MLM) and next sentence prediction (NSP) based on a significant quantity of text corpora. MLM is the task of predicting some randomly masked tokens, while NSP is to judge whether two sentences are paired. The pre-trained process makes language models to gain generic language representation and general knowledge. Previous studies have demonstrated that pre-trained language models can learn general knowledge

similar to traditional information extraction methods [15]. Further, it is proved that Prompt is an effective tool to activate the hidden knowledge when pre-trained language models are integrated into some specific downstream tasks [10].

In the early days, the pre-trained language model, GPT-1 [16], firstly attempts to utilize Prompt paradigm to solve the task of Sentiment Analysis. With the introduction of GPT-3 [17], fine-tuning based on the original model becomes dramatically more complex due to the vast number of parameters. Gradually, Prompt paradigm, a method that can utilize pre-trained language models via templates, is gaining popularity. The core idea is to transform the downstream task and adapt it to the pre-trained language model. Recently, Prompt has been applied in various NLP tasks such as text classification [11], [18], named entity recognition [19], [20], relation extraction [21], etc. The implementation of Prompt is becoming more diverse, with methods that freeze the entire parameters of PLMs [22], [23] and methods combining fine-tuning with Prompt [11], [24]. It is worth mentioning that previous research has discovered that when the parameters of pre-trained language models are entirely frozen, pre-trained language models with a higher number of parameters have a superior impact [23]. However, with fewer parameters, the combination of fine-tuning pre-trained language models with Prompt paradigm will perform better. An additional study [10] has also demonstrated that the method using Prompt paradigm can enhance the efficiency of sample utilization in low-resource settings, such an approach can be used to make models more suitable for zero-shot tasks. Therefore, this paper attempts to apply Prompt paradigm to zero-shot relation extraction by combining Prompt paradigm with fine-tuning PLMs.

Moreover, because the template is one essential aspect in Prompt, the related studies on it have never stopped. There are roughly three forms of template generation, that is, manual templates [17], [25], [26], discrete templates [18], [27], [28], [29] and continuous templates [11], [22], [24], [30]. The manual template is easier and less complicated. Generally, the template is designed according to prior expertise. Therefore, the method reduces dependency on supervised data and supports low-resource learning tasks better. But, there are obviously limitations, such as high cost, time consumption, and even template design being difficult in some cases. The discrete template in Prompt refers to the automated production of Prompt from natural language that humans can understand. Such a solution also eliminates the dependency on expert intervention by automatically selecting suitable templates. However, when using gradient-based back-propagation technique to optimize the model, the method will run into problems due to it not being able to deduce. This problem can be solved using an approximation methodology, which may lead to a detrimental influence on the overall performance. The continuous template in Prompt is no longer confined to plain language but also directly creates the embedding, allowing the whole model to be updated and improved through gradient back-propagation. This procedure, however, needs a certain quantity of data in order to exert its effectiveness, making it unsuitable for low-resource applications. To address the above issues, this paper proposes the method of dynamic template in Prompt, which adopts manual initialization with continuous template

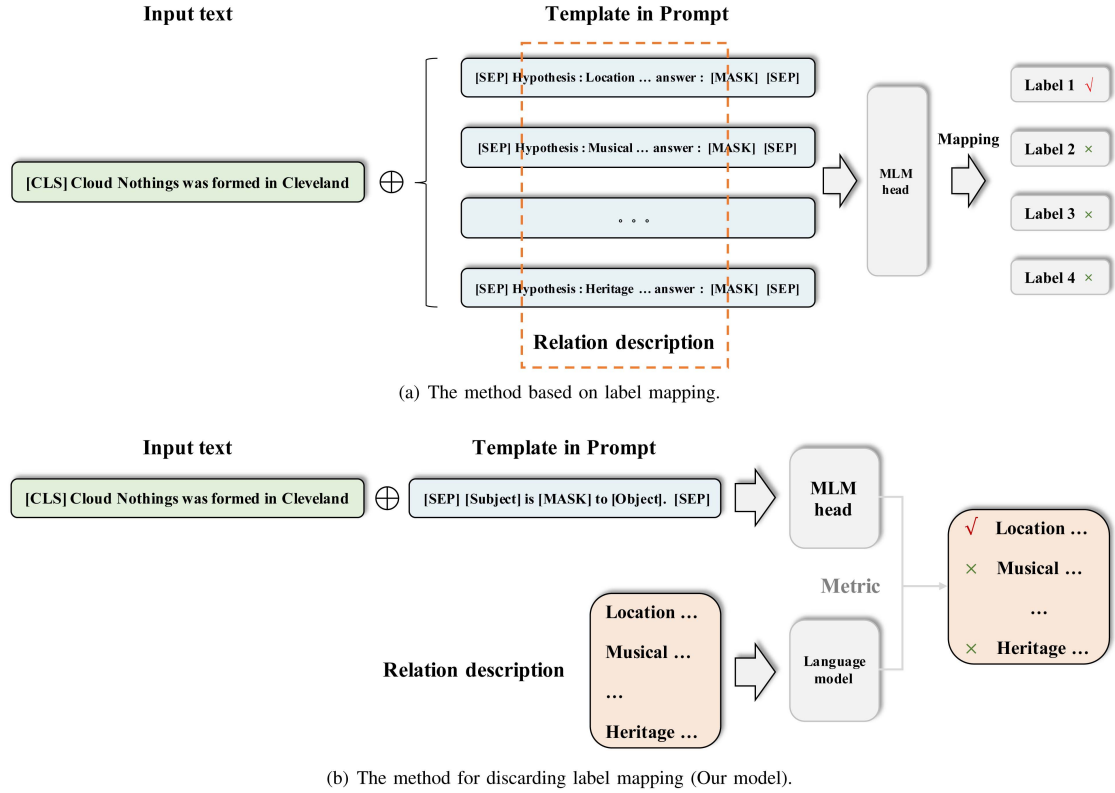


Fig. 1. Flowchart of Prompt-driven zero-shot relation extraction. (a) The generic and typical label mapping-based method. (b) The method of directly using [MASK] embedding to act as the generated relation representation.

and reduces the search space of Prompt-driven optimization to mitigate the data requirement and achieve a better performance in the task of zero-shot relation extraction.

B. Zero-Shot Relation Extraction

The concept of zero-shot learning involves establishing an intermediate semantic space that connects both seen and unseen classes [31]. In the context of relation extraction, zero-shot learning is utilized to recognize relations that were not seen during training. Early studies on zero-shot relation extraction typically transformed it into a new task, such as question-answering (QA) [6] and textual entailment [7], to fit the requirements of zero-shot learning. However, these approaches could not provide satisfactory semantic representation for both the seen and unseen relations. Fortunately, the development of PLMs, such as BERT [8], has significantly improved text representation. In recent studies, ZS-BERT [9] used BERT to map the sentence and relation description to the same semantic space, albeit through different mapping functions. Furthermore, IDZR [32] and ZRCM [33] respectively use discriminative embedding learning and contrastive learning to optimize the mapping of sentences and relations, thereby improving the accuracy of the model. Despite its promising results, these methods are limited by the conflicts between the specific task and the knowledge contained in PLMs.

Currently, there exist Prompt-based methods for the zero-shot relation extraction task, such as ZS-SKA [12], EPZR [34],

and RelationPrompt [13]. Intuitively, RelationPrompt models the zero-shot relation extraction task as a sequence generation task and then integrates Prompt to help sequence generation. Therefore, this approach deviates from the traditional Prompt paradigm. Fig. 1(a) illustrates that more representative Prompt paradigm method, such as EPZR, maps [MASK] tokens output by pre-trained language models to labels and then categorizes them by comparing the labels. Additionally, ZS-SKA employs a knowledge graph to aid in label mapping. The effectiveness of applying Prompt paradigm is closely linked to the appropriate mapping strategy, and an inappropriate mapping method can result in degraded performance. Furthermore, training the mapping function requires a certain amount of data, and the label cannot correspond to the unseen relation. In this paper, as depicted in Fig. 1(b), we deviate from the conventional approach of label mapping in Prompt paradigm and directly treat the embedding space of relation descriptions as the relation representation space. Prompt is utilized to determine the sentence representation via a task-specific template, and the sentence representation is ultimately compared with the relation representation in the shared semantic space. This paper presents a series of preliminary studies on Prompt-driven zero-shot relation extraction based on this approach.

III. PRELIMINARY STUDIES

In this section, we perform some tentative explorations for introducing Prompt paradigm to the task of zero-shot relation

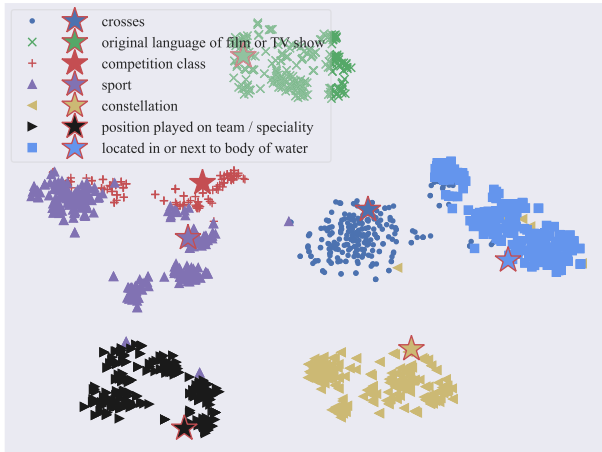


Fig. 2. Visualizing the generated vectors of relation description and [MASK] embedding of the template in FewRel dataset via *t-sne* dimension reductions. The template is “It can be seen that [Subject] is [MASK] to [Object]”. Different colors correspond to different relations. A pentagram of the same color indicates generated vectors of relation description.

extraction. An initial model following the framework shown in Fig. 1(b) has been implemented. Further, three preliminary studies are conducted to briefly validate the feasibility of the key idea in Fig. 1(b) and some current problems over the approach. In this way, the key issues of Prompt-driven zero-shot relation extraction are revealed.

A. Can the Embedding Generated by the Pre-Trained Language Model Be Qualified as the Representation Space of Relations?

Firstly, we find an interesting phenomenon when exploring Prompt paradigm in the task of relation extraction. In order to construct the relation extraction task as a MLM task, we integrate the sentence with a template within the Prompt, encompassing both the subject and object. The resulting combined text then serves as the input for pre-trained language models. As the pre-trained language model predicts, tokens at [MASK] location are similar to those in the relation description tokens labeled within the dataset. Taking a sentence in FewRel as an example, we combine the sentence and template in Prompt to create the input sequence, that is, “[CLS] In June 1987, the Missouri Highway and Transportation Department approved the design location of a new four-lane Mississippi River bridge to replace the deteriorating Cape Girardeau Bridge. [SEP] The Cape Girardeau Bridge [MASK] the Mississippi River. [SEP]”. This example is input to the bert-base-uncased model, and then the output is the word “crosses” at [MASK] position. The output is precisely the same as the label within the relation description. This phenomenon demonstrates that the embedding generated by the pre-trained language model may directly act as the relation representation. To confirm the above conjecture, we employ the pre-trained language model to convert relation description into embedding and visualize their dimension reductions via *t-sne*. As shown in Fig. 2, we randomly select some different classes of the relation description and [MASK] embedding of the template as examples. Diverse colors are associated with distinctive relations,

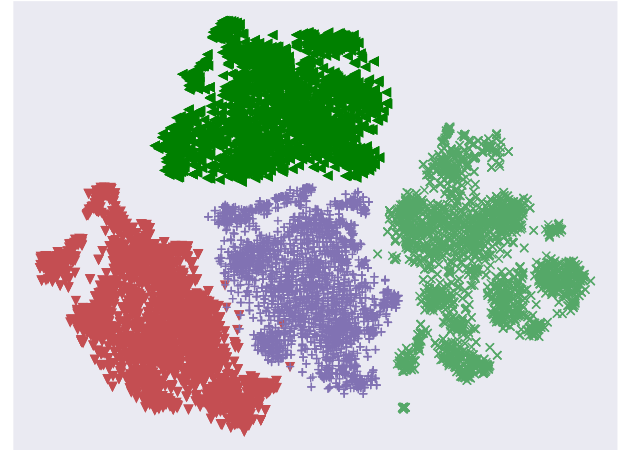


Fig. 3. Visualizing the generated [MASK] embedding by Prompt-based MLM with different templates for FewRel dataset via *t-sne* dimension reductions. Different colors and markers indicate [MASK] embedding vectors with different templates. These templates are: (a) “It can be seen that [Subject] is [MASK] to [Object]”, (b) “As can be seen from the above, the relationship between [Subject] and [Object] is [MASK]”, (c) “The relation of [Subject] and [Object] is [MASK]”, (d) “Obviously, [MASK] is the relationship between [Subject] and [Object]”.

whereas a pentagon of identical color denotes the generated vectors of relation depiction. It can be seen that an embedding mingles others with the same relation classes while embeddings of different relation classes lead to larger inter-relation distance. The [MASK] embedding has shown a tendency to be closer to the generated vectors of relation description. Therefore, in this paper, we directly regard the embeddings generated by the pre-trained language model as the relation representation.

B. Does the Template in Prompt Affect the Representation Space of Relations?

The appropriate method for constructing the templates in Prompt is one vital issue when applying Prompt paradigm in other tasks. In this preliminary study, we attempt to explore the degree of impact of the templates in relation extraction task. Four different templates in Prompt are manually constructed and further filled with sentences from the FewRel dataset with corresponding subjects and objects. Then, we visualize the [MASK] embedding generated by the PLMs using *t-sne* dimension reductions. As shown in Fig. 3, there is a clear difference among the embeddings with four different templates in Prompt. Such a phenomenon indicates that the templates severely impact the ability of the relation representation in the model based on Prompt paradigm. Indeed, the phenomenon is easy to interpret when relation extraction is implemented based on a pre-training MLM task, which trains the language models to mask out 15% of the tokens in the sentences and then predicts the masked tokens. Furthermore, the template type tremendously impacts this task’s effect, and different templates in Prompt can determine different attributes of the tokens predicted by the pre-trained language model. For example, the pre-trained model predicts a higher probability that the tokens are adjectives with the template whose [MASK] tokens are in the position of adjective. In summary, to

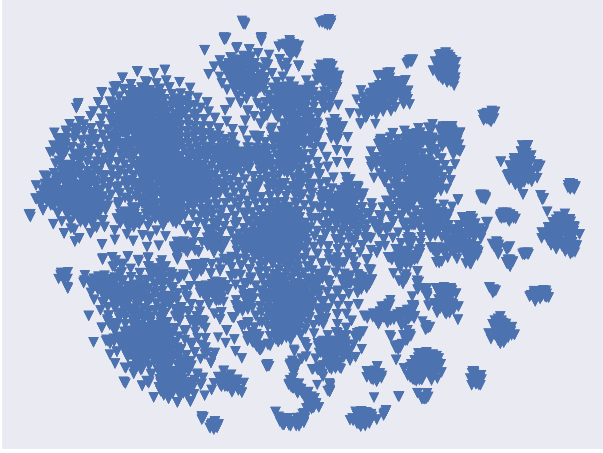


Fig. 4. Visualizing the generated [MASK] embedding in FewRel dataset with the frozen template via *t-sne* dimension reductions. Different templates are tried for the preliminary study and their distribution has a similar pattern. Limited by space, we randomly select one for display.

achieve a suitable relation representation, the choice of template is one vital issue. To solve it, in this paper, we propose the strategy of dynamic template in Prompt.

C. What is the Distribution of [MASK] Embeddings Generated by Prompt?

The last preliminary study investigates the factors that influence the use of Prompt paradigm for the task of relation extraction. As mentioned above, we combine the sentences from the testset of FewRel with templates and freeze the template in Prompt. Then, they are input into the model and visualize the generated vectors using *t-sne* dimension reductions. As shown in Fig. 4, it can be observed that the embeddings are clustered on the left side, whereas the right side is sparse. In brief, the embeddings have an uneven distribution. The problem associated with this phenomenon significantly affects the model's ability to generate the relation representation. Meanwhile, previous studies [35], [36], [37] mention that the embedding output by the pre-trained language model suffers from a representation degradation problem, with anisotropic trends in the vector distribution. The representation degradation problem leads to the phenomenon that most of pre-trained language models output conical distribution in the embedding space [38]. This problem is because the pre-trained language models use different frequency tokens in the corpus during the pre-training stage. The fundamental idea of the Prompt paradigm applied to the relation extraction zero-shot task is to fully use the knowledge learned by the pre-trained language model. In summary, the representation degradation problem can lead to the generated relation representations not guaranteeing alignment and uniformity. This paper attempts to solve the problem by fusing the key idea of fine-tuning with contrastive learning and Prompt paradigm.

IV. THE PROPOSED DYNAMIC PROMPT-DRIVEN MODEL

Given a subject s , an object o and their context sentence $x = [x_1, x_2, \dots, x_{|x|}]$ where s and o are two n -gram phrases of

x , the task of relation extraction is to establish a model that can recognize the relation class between s and o indicated by x from a known relation set R . For the task of zero-shot relation extraction, the model has to recognize unseen relations, that is, $R = R_{train} \cup R_{test}$ and $R_{train} \cap R_{test} = \emptyset$. Nevertheless, each relation $r_i \in R$ has a unique textual description, denoted as d_i . Under the background of such a task, we present the proposed dynamic Prompt-driven model in this section.

A. The Overall Framework

Fig. 5 presents the overall framework of the proposed dynamic Prompt-driven model for zero-shot relation extraction. Based on PLMs and relation descriptions, it is easy to obtain the representation for one specific relation r_i , denoted as $\mathbf{r}_i \in \mathbb{R}^b$. b is the embedding dimension of PLMs. Thus, the key to the model lies in generating the context representation, denoted as $\tilde{\mathbf{r}} \in \mathbb{R}^b$, which can be compared with the relation representation. Considering the small number of training samples, fine-tuning PLMs is challenging. Therefore, the relation representation $\mathbf{r}_i$ is obtained by max-pooling the embeddings of each word in d_i , instead of using the embedding of [CLS] token. Meanwhile, as above-mentioned, [MASK] embedding from Prompt-based MLM is assigned as the context representation $\tilde{\mathbf{r}}$. The optimization goal of the model is to minimize the distances between $\tilde{\mathbf{r}}$ and the representations of its corresponding relations, as (1).

$$\mathcal{L}_{\mathcal{R}} = \sum_j \sum_i^{|R_{train}|} k_{ij} \cdot f_{distance}(\tilde{\mathbf{r}}_j, \mathbf{r}_i), \quad (1)$$

where $k_{ij} = 1$ if the j -th training sample belongs to the i -th relation, otherwise $k_{ij} = 0$. $f_{distance}(\cdot)$ is a distance measure function, which is implemented by Euclidean distance.

B. Prompt-Based MLM

The goal of Prompt-based MLM is to generate the context representation, where Prompt paradigm is utilized to stimulate the task-specific knowledge contained in PLMs. To explain the mechanism of Prompt-based MLM, we first denote the template in Prompt as $T(s, o)$. As shown in (2), for the task of relation extraction, a template consists of a pre-defined token sequence and the spaces reserved for the subject and object, where p and q represent the corresponding positions. t denotes a token in the template. When a template is used, the spaces will be filled by the given subject s and object o .

$$T(s, o) = [t_1, \dots, t_p] s [t_{p+1}, \dots, t_q] o [t_{q+1}, \dots, t_{|T(s, o)|}], \quad (2)$$

As shown in (3), for a specific input (x, s, o) , the input of Prompt-based MLM, denoted as $\tilde{x}$, is obtained by filling the template and concatenating it with x via [CLS] and [SEP] tokens.

$$\tilde{x} = [\text{CLS}] x [\text{SEP}] T(s, o) [\text{SEP}], \quad (3)$$

Based on this input format, a general masked language model usually makes a word-distribution prediction for the position of the [MASK] token, but the distribution cannot be directly compared with the relation representation. Since the first preliminary study indicates that [MASK] embedding can be qualified

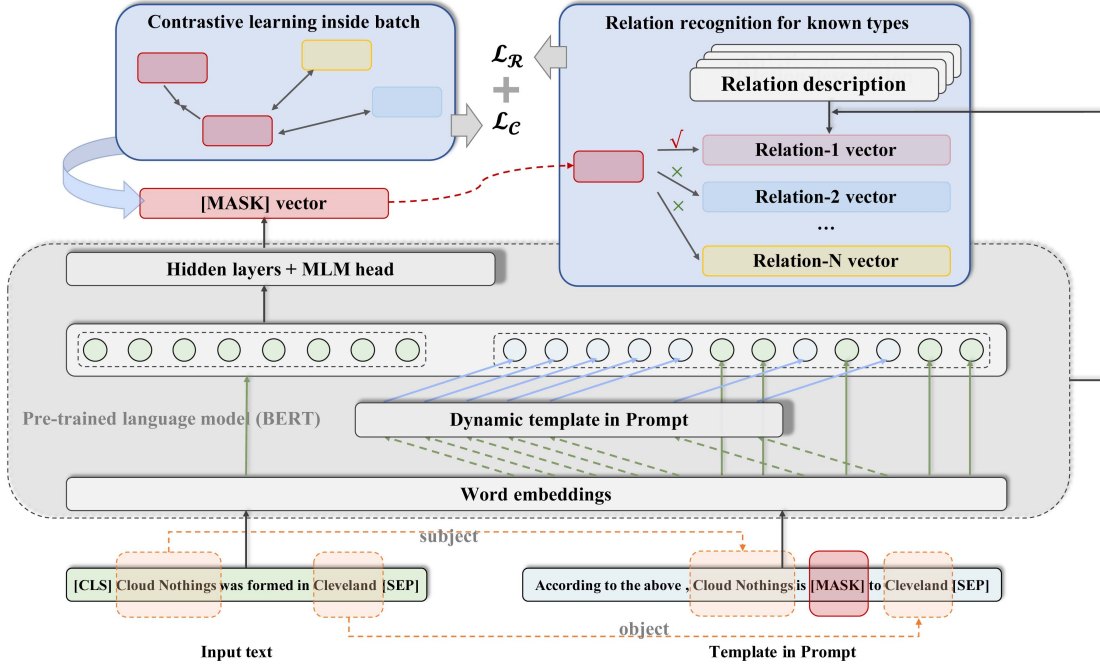


Fig. 5. The framework of **Dynamic Prompt-Driven Model (DPDM)** for zero-shot relation extraction. Compared with the traditional Prompt paradigm, the method proposes the dynamic template in Prompt and combines it with prototype learning for zero-shot relation extraction. Meanwhile, contrastive learning strategies are employed to optimize the representation of relations.

as the relation representation, in the proposed model, we assign [MASK] embedding as $\tilde{\mathbf{r}}$. Referring to the details in BERT [8], $\tilde{\mathbf{r}}$ is obtained as (4).

$$\tilde{\mathbf{r}} = \text{LayerNorm}(\text{Dropout}(\hat{\mathbf{e}}_t + \hat{\mathbf{e}}_p + \hat{\mathbf{e}}_s)), \quad (4)$$

where $\hat{\mathbf{e}}_t, \hat{\mathbf{e}}_p, \hat{\mathbf{e}}_s \in \mathbb{R}^b$ respectively represent the token embedding, position embedding and segment embedding of [MASK] in the last layer of PLM.

C. Dynamic Template in Prompt

Aimed at adapting to the task of relation extraction, the initial template $T(s, o)$ is defined as “{subject} is [MASK] to {object}. [SEP]”. The corresponding embedding $\mathbf{e}(T(s, o)) \in \mathbb{R}^{|T(s, o)| \times b}$ is denoted as (5) without dynamic transformation,

$$h_{\oplus}(\mathbf{e}(s), \mathbf{e}(is), \mathbf{e}([\text{MASK}]), \mathbf{e}(to), \mathbf{e}(o), \mathbf{e}(\cdot), \mathbf{e}([\text{SEP}])), \quad (5)$$

where $\mathbf{e}(\cdot)$ is the layer of word embeddings and $h_{\oplus}(\cdot)$ is the concatenation operation. In general, the embeddings that have been concatenated are input into the following layers of BERT, which include self-attention and feed-forward neural networks. To further improve the flexibility, a strategy of dynamic template in Prompt is proposed for zero-shot relation extraction. For the tokens in the template, the word embeddings are not directly input into the following layers but are allowed to be transformed at a certain scale. Concretely, an extra vector corresponding to $T(s, o)$ is introduced, denoted as $\mathbf{M} = [m_s, m_{is}, m_{[\text{MASK}]}, m_{to}, m_o, m_{\cdot}, m_{[\text{SEP}]}] \in \mathbb{R}^{|T(s, o)| \times 1}$. Among the elements of $\mathbf{M}$, $m_s, m_{[\text{MASK}]}, m_o$ and $m_{[\text{SEP}]}$ are always equal to 1, and the rest are trainable. When the dynamic

template in Prompt is applied, the embedding $\mathbf{e}(T(s, o))$ is updated to $\tilde{\mathbf{e}}(T(s, o))$ via (6).

$$\tilde{\mathbf{e}} = \mathbf{M} \otimes \mathbf{e}, \quad (6)$$

where $\otimes$ is the element-level multiplication operation, e.g. $\tilde{\mathbf{e}}(is) = m_{is} \cdot \mathbf{e}(is)$. In this case, the embeddings of tokens in the template achieve an update. The update process is independent of PLMs, leading to the dynamic template.

D. Contrastive Learning for Representation

To solve the problem of representation degeneration, we introduce the idea of contrastive learning during the training. The key idea is to reduce the distance between samples with the same relation. Correspondingly, the distance between different relations should be increased. We introduce contrastive learning into each batch of training data. Equation (7) presents the optimization goal of this component.

$$\mathcal{L}_C = \sum_k -\log \frac{e^{g_{\text{distance}}(\tilde{\mathbf{r}}_k, \tilde{\mathbf{r}}'_k)/\tau}}{\sum_l^{batch} (e^{g_{\text{distance}}(\tilde{\mathbf{r}}_k, \tilde{\mathbf{r}}'_l)/\tau})}, \tau = 0.05, \quad (7)$$

where $g_{\text{distance}}(\cdot)$ is a distance measure function implemented by Cosine distance. Following an unsupervised strategy [39], for each sample, its [MASK] representation $\tilde{\mathbf{r}}_k$ will be compared with that of other samples in the same training batch and deformation of itself, that is, $\tilde{\mathbf{r}}_l$ and $\tilde{\mathbf{r}}'_k$. $\tilde{\mathbf{r}}'_k$ is obtained via a dropout layer on top of $\tilde{\mathbf{r}}_k$. It is worth mentioning that, to facilitate implementation, we ensure that the samples in the same batch have different relations. Finally, the overall model is trained by combining $\mathcal{L}_R$ and $\mathcal{L}_C$.

TABLE I
STATISTICS OF THE DATASETS

Datasets	FewRel	Wiki-ZSL	TACRED
Relations	80	113	42
Samples	56,000	94,383	106,264
Average sentence length	25.0	24.9	36.4
Train relations	65	98	30
Test relations	15	15	11

V. EXPERIMENTS

To verify the effectiveness of the proposed model, a series of experiments are conducted to answer the following questions:

- *RQ-1*: How does the proposed DPDM model perform compared with the existing advanced baselines?
- *RQ-2*: Is the proposed DPDM model robust enough under different data conditions?
- *RQ-3*: How does the proposed dynamic template in DPDM model perform compared with the existing templates?
- *RQ-4*: How do the components (dynamic template and contrastive learning) affect the proposed DPDM model?
- *RQ-5*: Has the representation degradation problem been alleviated by the proposed DPDM model?

A. Experimental Settings

1) *Datasets*: We conduct extensive experiments on three benchmark datasets, that is, FewRel [40], Wiki-ZSL [9] and TACRED [41]. FewRel is a dataset manually annotated for few-shot relation extraction task, which contains 80 relations, each relation has 700 samples and their detailed descriptions. Wiki-ZSL is a dataset constructed by remote supervision of Wikipedia articles and Wiki-knowledge bases. It contains 113 different classes with a total of 94383 samples. TACRED is a human-annotated relation extraction dataset containing 106,264 samples and encompassing 42 relations. Additionally, following the experimental settings of [34], we removed the “no_relation” relation from the TACRED dataset and conducted experiments on the remaining 41 categories. Moreover, relations in TACRED are not described, so we have supplemented this information with the *TAC KBP 2015 Slot Descriptions*¹ document. Table I provides an overview of the dataset statistics that were analyzed in this study. For the zero-shot relation extraction task, we utilized the seen relations for training, while retaining the unseen relations for testing. To ensure consistency with prior research [13], [32], [33], [34], we employed a methodology of randomly categorizing all relations into either seen or unseen classes. We repeated the experiment five times and reported the average of the results.

2) *Baselines*: We compare the proposed model with various baselines from several perspectives. **DPDM** refers to the proposed model introducing dynamic templates and contrastive learning, while **PDM** is the basic model following the idea in Fig. 1(b). We extend previous taxonomy [9] and divide the baselines into four classes: (i) supervised learning models,

(ii) textual entailment models, (iii) Representation learning models and (iv) Prompt-based models. In supervised relation extraction models, referring to previous research [9], we change the last hidden layer to a fully connected layer with *tanh* activation, and the output of the vector from this layer is treated as relation representations. The textual entailment model formulates the zero-shot relation extraction task as a textual entailment task. They take sentences and each possible relation type as input to the classification task. The determination of what relation is in the sentence is made by judging whether the label is a semantic match to the sentence.

(i) Supervised learning models

- *Bi-LSTM* is composed of forward LSTM and backward LSTM, which solves the memory problem of RNN (Recurrent Neural Network) and is able to capture long-distant text information [42].
- *Att Bi-LSTM* leverages attention mechanism into Bi-LSTM and thus can capture more important semantic information in a sentence [3].
- *R-BERT* is a supervised model for relation classification. To make it more suitable for zero-shot relation extraction, following the previous practice [9], we use the sentence representation to perform the nearest neighbor search on label embeddings [43].

(ii) Textual entailment models

- *ESIM* is a semantic matching model that uses Bi-LSTM for encoding, and then the local inference model and softmax are used as the decoding layer [44].
- *CIM* replaces Bi-LSTM with a conditional encoder. The relation description is input to the encoder along with the sentence and decoded by binary classification for each possible relation.

(iii) Representation learning models

- *ZS-BERT* converts sentences into sentence representations and leverages the textual relation descriptions to generate relation representations. It matches the final results using a nearest neighbor search on the candidate relation representations [9].
- *ZSCM* introduces contrastive learning on the basis of ZS-BERT to improve the ability for mining hidden text information. [33]
- *IDZR* proposes a new model that incorporates discriminative embedding learning to optimize the generation of sentence and semantic relation representations. [32]

(iv) Prompt-based models

- *ZS-SKA* is a relation extraction model that utilizes Prompt paradigm and external knowledge graphs to enhance its semantic capabilities. This is achieved through a prototype network that generates enhanced relation representations for both seen and unseen relations. Finally, the model uses a distance metric to perform the final matching [12].
- *EPZR* method concatenates sentences with textual descriptions of each type that is to be extracted. This approach leverages Prompt paradigm to transform zero-shot relation extraction methods into textual entailment tasks [34].
- *RelationPrompt* learns a text generation model by using the training samples of the seen relations. Then, these models are utilized to generate the training samples of

¹[Online]. Available: https://tac.nist.gov/2015/KBP/ColdStart/guidelines/TAC_KBP_2015_Slot_Descriptions_V1.0.pdf

TABLE II
PERFORMANCE COMPARISON ON FEWEREL DATASETS WITH DIFFERENT NUMBER OF TRAINING SAMPLES (%)

Model	n=20			n=100			n=200			n=all		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
Bi-LSTM	—	—	—	—	—	—	—	—	—	16.83	27.62	20.92
Att Bi-LSTM	14.19	13.88	14.03	15.75	19.8	17.55	20.83	26.00	23.13	38.13	32.05	34.82
R-BERT	8.40	8.38	8.39	13.61	15.90	14.67	16.05	18.58	17.22	32.25	25.58	28.53
CIM	—	—	—	—	—	—	—	—	—	31.83	33.06	32.43
ESIM	0.60	5.45	1.08	0.90	6.56	1.58	7.66	7.38	7.52	36.97	32.51	34.60
ZS-BERT	6.04	6.36	6.20	6.34	7.93	7.05	8.35	9.59	8.93	35.54	38.19	36.82
IDZR	—	—	—	—	—	—	—	—	—	43.96	39.11	41.36
ZSCM	—	—	—	—	—	—	—	—	—	40.27	40.72	40.50
ZS-SKA	—	—	—	—	—	—	—	—	—	45.34	51.67	47.02
EPZR	44.94	45.72	45.33	50.21	50.72	50.96	52.49	53.98	53.23	64.48	62.45	63.45
RelationPrompt	17.62	48.33	25.84	53.90	49.19	51.44	56.59	50.06	53.12	74.33	72.51	73.40
PDM (ours)	45.01	49.31	47.06	58.06	63.73	60.76	63.30	66.29	64.76	69.85	72.74	71.27
DPDM (ours)	47.39	63.08	54.12	65.62	73.54	69.35	71.36	77.10	74.12	76.52	77.57	77.04

TABLE III
PERFORMANCE COMPARISON ON TACRED DATASETS WITH DIFFERENT NUMBER OF TRAINING SAMPLES (%)

Model	n=20			n=100			n=200			n=all		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
Att Bi-LSTM	14.33	11.38	12.68	13.73	10.64	11.99	15.68	21.70	18.20	25.20	20.17	22.41
R-BERT	14.59	7.27	9.70	18.93	12.12	14.78	23.62	19.67	21.47	44.66	45.86	45.25
ESIM	9.09	0.15	0.29	8.54	9.41	8.96	1.52	9.15	2.61	26.99	18.38	21.87
ZS-BERT	10.79	9.35	10.02	12.53	9.25	10.64	14.98	15.79	15.38	38.08	42.72	40.27
EPZR	32.40	30.54	31.44	38.12	22.75	28.50	34.56	33.73	34.14	51.85	44.71	48.03
RelationPrompt	0.00	0.00	0.00	18.54	60.01	28.33	27.25	68.55	39.00	35.64	68.56	46.90
PDM (ours)	27.36	30.99	29.06	45.83	48.71	47.23	50.42	56.40	53.24	59.64	63.33	61.43
DPDM (ours)	35.36	38.02	36.64	57.48	59.38	58.41	70.03	73.80	72.90	77.21	78.58	77.89

the unseen relations. The zero-shot relation extraction task is subtly transformed into a supervised relation extraction task through an iterative approach [13].

3) *Implementation and Evaluation Metrics*: We implement the models using PyTorch [45] and train them on a 2080Ti GPU. The uncased-BERT-base model provided by Huggingface's Transformers² library is adopted as the encoder for fair comparison. AdamW [46] optimizer is applied with a learning rate of $2e-5$. Because contrastive learning is employed, we ensure that the samples in one batch belong to different relations during training. Consistent with previous studies, the performance is evaluated by precision (Prec.), recall (Rec.), and F1 score (F1).

B. Performance Comparison (RQ-1)

To verify the effectiveness of the proposed model, there are two cases on performance comparison following previous studies. The first is the one of rich relation classes and low data resources, which is conducted on FewRel and TACRED with different number of training samples (n). The other is the one of few unseen relation classes and sufficient data resources, which is conducted on Wiki-ZSL and FewRel with different numbers of unseen relations (m). The details are as follows.

1) *The Case of Rich Relation Classes and Low Data Resources*: Table II and Table III show the performance comparison in the case of rich relation classes and low data resources. In detail, we explore the performance of the proposed model via freezing m and modifying n . The presented table provides evidence that prompt-driven methods outperform representation

learning-based models in zero-shot learning tasks. Specifically, when $n = \text{all}$, the performance of the representation learning-based method is generally lower than that of the prompt-driven method. Additionally, the performance of representation learning-based method significantly decays as n decreases, while the prompt-driven model maintains stable and superior performance due to its ability to leverage task-specific knowledge in the pre-trained language model. Notably, the proposed method is optimized using the prompt paradigm and achieves the best results compared to other prompt-based methods, validating its effectiveness. Traditional prompt-driven methods, such as ZS-SKA and EPZR, rely on label mapping to optimize the model's performance. In contrast, DPDM achieves significant improvements by exploiting the benefits of Prompt paradigm without relying on label mapping. Additionally, RelationPrompt performs well, but it faces increased data requirements due to the larger model size and training datasets of its pre-trained models, GPT-2 and BART. Moreover, in the experimental setting with $n=20$ in the TACRED dataset, RelationPrompt encounters difficulty in generating enough training data for an unseen relation extraction model. Conversely, DPDM optimizes the prompt paradigm while addressing its limitations and maintaining its advantages, leading to superior performance.

2) *The Case of Few Unseen Relation Classes and Sufficient Data Resources*: Table IV shows the performance comparison in the case of a few unseen relation classes and sufficient data resources. In detail, we explore the performance of the proposed model via freezing $n = \text{all}$ and modifying m . The Prompt-driven method exhibits superior performance compared to other methods, particularly in the most challenging recognition task when $m = 15$, and DPDM achieves the best results. Notably, reducing the number of unseen relation classes for representation learning

²[Online]. Available: <https://github.com/huggingface/transformers>

TABLE IV
PERFORMANCE COMPARISON ON WIKI-ZSL AND FEWEREL DATASETS WITH DIFFERENT NUMBERS OF UNSEEN RELATIONS (%)

Unseen labels	Model	Wiki-ZSL			FewRel		
		Prec.	Rec.	F1	Prec.	Rec.	F1
m=5	R-BERT	39.22	43.27	41.15	42.19	48.61	45.17
	CIM	49.63	48.81	49.22	58.05	61.92	59.92
	ZS-BERT	71.54	72.39	71.96	76.96	78.86	77.90
	IDZR	87.48	77.50	82.19	87.11	86.29	86.69
	ZSCM	76.15	77.10	76.60	86.70	84.51	85.60
	RelationPrompt	70.66	83.75	76.63	90.15	88.50	89.30
	DPDM	84.67	92.56	87.43	90.08	96.43	93.15
m=10	R-BERT	26.18	29.69	27.82	25.52	33.02	28.20
	CIM	46.54	47.90	45.57	47.39	49.11	48.23
	ZS-BERT	60.51	60.98	60.74	56.92	57.59	57.25
	IDZR	71.59	64.69	67.94	64.41	62.61	63.50
	ZSCM	62.41	64.16	63.27	53.67	53.96	53.81
	RelationPrompt	68.51	74.76	71.50	80.33	79.62	79.96
	DPDM	84.67	84.96	84.82	85.48	88.58	87.00
m=15	R-BERT	17.31	18.82	18.03	16.95	19.37	18.08
	CIM	29.17	30.58	29.86	31.83	33.06	32.43
	ZS-BERT	34.12	34.38	34.25	35.54	38.19	36.82
	IDZR	38.37	36.05	37.17	43.96	39.11	41.36
	ZSCM	33.47	36.71	35.01	40.27	40.72	40.50
	ZS-SKA	41.03	40.12	38.13	45.34	51.67	47.02
	EPZR	-	-	-	64.48	62.45	63.45
	RelationPrompt	63.69	67.93	65.74	74.33	72.51	73.40
	DPDM	79.41	80.03	79.72	76.52	77.57	77.04

TABLE V
PERFORMANCE COMPARISON ON CROSS-DOMAIN TASKS (%)

Model	FewRel to TACRED			TACRED to FewRel		
	Prec.	Rec.	F1	Prec.	Rec.	F1
Att Bi-LSTM	21.86	27.72	24.44	31.27	39.26	34.82
ESIM	22.67	18.91	20.62	19.38	11.93	14.77
R-BERT	23.10	28.49	23.98	15.31	14.70	15.00
ZS-BERT	35.90	29.78	32.55	17.69	11.81	14.16
EPZR	41.26	37.62	39.36	60.01	50.74	54.99
PDM (ours)	50.25	57.56	53.66	41.78	46.38	43.96
DPDM (ours)	60.69	66.01	63.24	52.31	61.09	56.36

TABLE VI
PERFORMANCE COMPARISON BETWEEN DIFFERENT TEMPLATES AND DYNAMIC TEMPLATE (%)

	FewRel			TACRED			Wiki-ZSL		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
T1	68.15	71.32	69.70	49.66	55.17	52.27	69.78	70.66	70.22
T2	67.18	70.16	68.64	57.43	61.51	59.40	70.38	71.21	70.79
T3	68.95	70.42	69.68	54.71	59.47	56.99	71.07	73.32	72.18
T4	68.47	71.44	69.92	47.36	53.22	50.12	69.91	71.23	70.56
T5	68.71	71.34	69.99	59.10	63.11	61.04	70.92	71.37	71.14
T6	66.66	71.27	70.47	48.25	54.83	51.33	66.77	64.49	65.61
T7	69.85	72.74	71.27	56.32	60.14	58.17	70.49	71.87	71.18
CT	69.09	70.51	67.72	55.05	59.90	57.37	67.23	65.62	68.91
DT	70.39	76.15	73.16	60.24	64.43	62.26	70.11	74.99	72.47

significantly enhances the model's performance. ChihYao [9] suggests that representation learning can effectively identify abstract attributes in input sentences, with this effect becoming increasingly pronounced as the extraction difficulty decreases. In this paper, we propose a method based on Prompt paradigm that

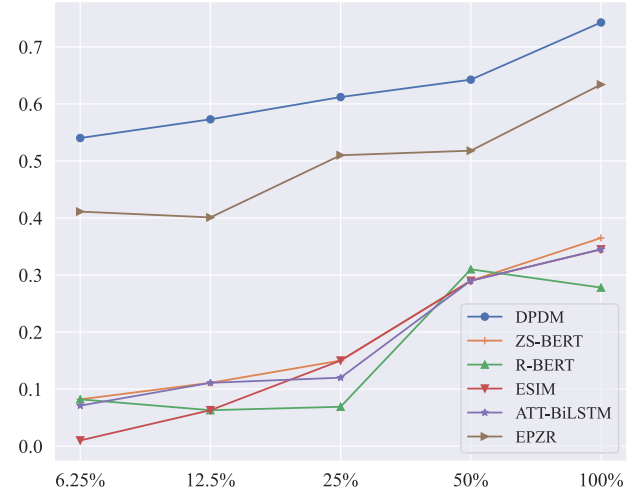


Fig. 6. Performance comparison with different numbers of seen relations in FewRel. The horizontal axis is the percentage of seen relations in total and the longitudinal axis is F1 score. As the number of seen relations decreases, DPDM maintains a high level of performance.

discards label mapping, converts the task into a representation generation task, and optimizes it using contrastive learning while retaining the features of representation learning. As the task difficulty decreases, our method achieves greater precision in relation extraction.

C. Robustness Study (RQ-2)

To illustrate the robustness of the proposed model, two experiments are conducted in this part. On one hand, we reduce the number of seen relations to train the model. On the other hand, we test the model outside the domain of training data.

1) *Different Numbers of Seen Relations*: In this experiment, we adjust the seen relations in FewRel, to obtain the training sets with different numbers of seen relations (i.e. 5, 9, 17, 33, 65). Then, the models trained by different training sets are compared on the original testing set. The result is shown in Fig. 6. Obviously, the prompt-driven approach, which effectively utilizes pre-training tasks, achieves stable performance and exhibits superior robustness. The proposed DPDM model exhibits a notable distinction from ZPER's approach of label mapping, specifically in the "match" and "not_match" words. Instead, the DPDM model discards this mapping process and compares labels directly in the word embedding space. As a result, leveraging the existing knowledge from pre-trained language models enables the model to achieve desirable performance even when trained on a limited number of seen relation classes. Overall, the performance of the model is positively correlated with the number of seen relation classes. As the number of seen relations decreases, the performance of each model decreases, but DPDM's decrease is smoother. With few seen relations, DPDM can also perform more satisfactorily than the baselines.

2) *Cross-Domain Studies*: As for robustness, an interesting question is whether the proposed DPDM model can recognize the relations of an unknown domain through the data of known

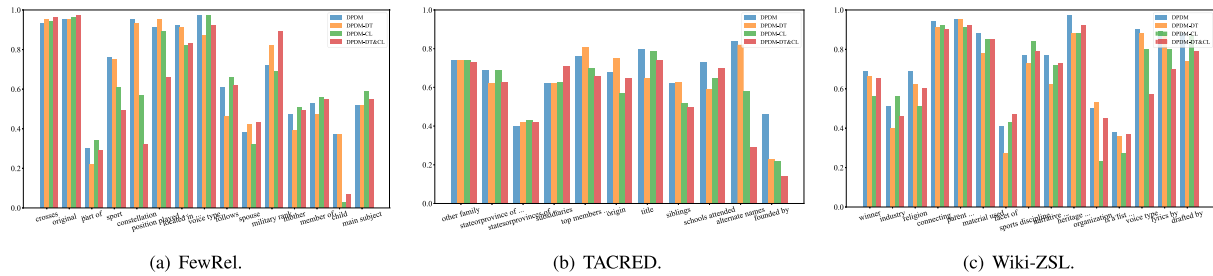


Fig. 7. Performance comparison over various relations in terms of F1 score. The horizontal axis is the name of unseen relations and the longitudinal axis is F1 score. It is demonstrated that the proposed DPDM model achieves a higher generalization ability at the slight expense of the performance of some well-performing relations.

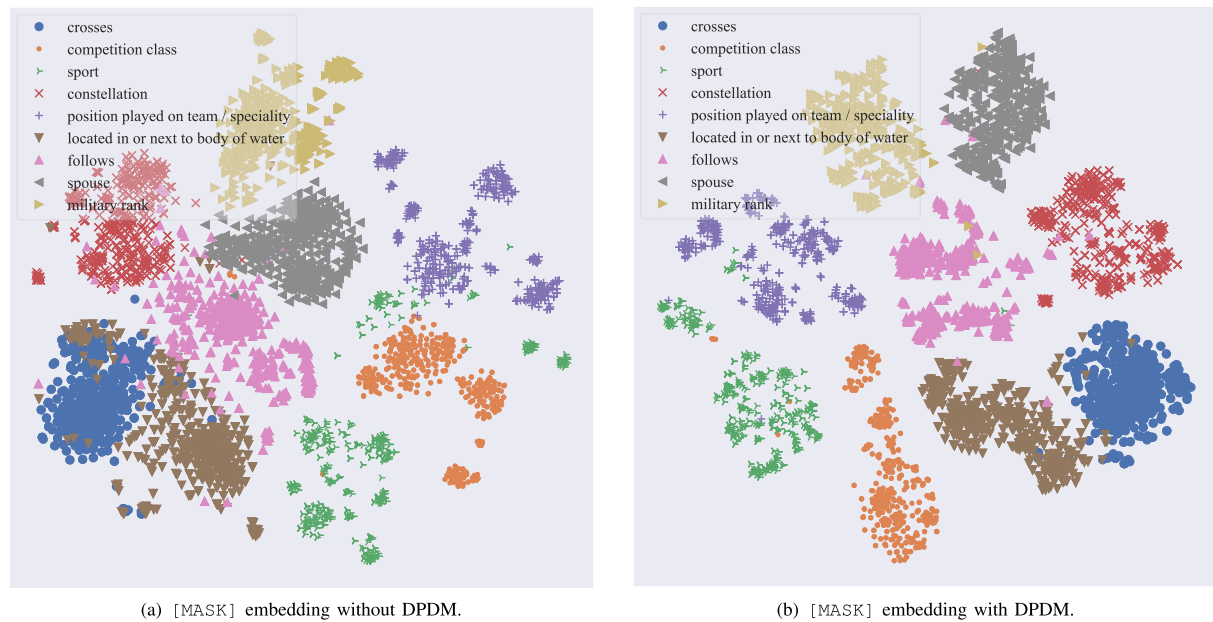


Fig. 8. Visualizing the generated [MASK] embedding of different relations in FewRel dataset via *t-sne* dimension reduction. It suggests that the representation degradation problem mentioned by the above preliminary study has been alleviated.

domains. To answer this question, following previous work [34], two tasks of cross-domain zero-shot relation extraction are conducted, that is, FewRel to TACRED and TACRED to FewRel. The results are presented in Table V. Obviously, compared with existing baselines, Prompt-driven model achieves a better generalization performance. In the case of using all the training samples, the proposed DPDM model also achieves state-of-the-art performance for cross-domain zero-shot relation extraction.

D. Effectiveness of Dynamic Template (RQ-3)

To further verify the effectiveness of dynamic template, in this experiment, we compare it with Prompt-driven models using some manually-design templates (from T1 to T7) as well as the strategy of the continuous template (CT). These models are displayed as follows:

- T1: As can be seen from the above, the relationship between [Subject] and [Object] is [MASK].
- T2: [Subject] [MASK] [Object].

- T3: [Subject] is [MASK] to [Object].
- T4: As seen above, the link between [Subject] and [Object] is [MASK].
- T5: It can be seen that [Subject] is [MASK] to [Object].
- T6: The relation of [Subject] and [Object] is [MASK].
- T7: Obviously, [MASK] is the relationship between [Subject] and [Object].
- CT: Referring to [24], some vectors of the same size as the token embedding are initialized and learned by gradient descent.

The experimental result is shown in Table VI, where DT denotes the proposed dynamic template method. It is demonstrated that the model with dynamic template performs better than the manually-design ones or continuous template. Meanwhile, performance improvement differs in different datasets. Compared with the baseline templates, dynamic template has a more significant improvement on FewRel and TACRED than

TABLE VII
ABLATION STUDY ON FEWREL, TACRED AND WIKI-ZSL (%). DT: DYNAMIC TEMPLATE

Datasets	Model	n=20			n=100			n=200			n=all		
		Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.
FewRel	DPDM	47.39	63.08	54.12	65.62	73.54	69.35	71.36	77.10	74.12	76.52	77.57	77.04
	-DT	46.26	50.14	48.12	64.37	72.08	68.01	69.75	75.62	72.57	74.51	77.16	75.81
	-CL	45.58	53.84	49.37	64.02	69.58	66.68	70.83	74.93	72.82	70.39	76.15	73.16
	-DPCL	45.01	49.31	47.06	58.06	63.73	60.76	63.30	66.29	64.76	69.85	72.74	71.27
TACRED	DPDM	35.36	38.02	36.64	57.48	59.38	58.41	72.03	73.80	72.90	77.21	78.58	77.89
	-DT	28.44	31.52	29.90	47.36	49.02	48.18	52.48	56.38	54.36	60.32	66.68	63.34
	-CL	28.52	31.67	30.01	49.21	49.77	49.48	53.41	57.49	55.37	60.24	64.43	62.26
	-DTCL	27.36	30.99	29.06	45.83	48.71	47.23	50.42	56.40	53.24	59.64	63.33	61.43
Wiki-ZSL	DPDM	39.05	40.74	39.89	57.21	65.83	64.57	74.28	74.08	74.19	79.41	80.03	79.72
	-DT	30.54	31.20	30.87	62.98	67.04	64.95	71.03	73.38	72.18	75.81	76.34	76.07
	-CL	38.34	39.59	39.00	61.06	66.71	63.76	69.30	71.28	70.28	70.11	74.99	72.47
	-DTCL	23.33	26.25	24.66	57.32	60.74	58.98	68.65	70.20	69.41	70.47	73.41	71.91

CL: contrastive learning. "-A" means we remove the components from the original DPDM.

Wiki-ZSL. In essence, dynamic template can be regarded as one optimization solution, which makes a trade-off between the large trainable parameters and the dynamicity of templates in Prompt. Such a strategy is able to alleviate the requirement of large-scale datasets and achieves a better performance in the task of zero-shot relation extraction.

E. Ablation Study (RQ-4)

To get deep insights into the impact of different components (i.e. dynamic template and contrastive learning), we conducted ablation studies on the three datasets mentioned above by removing each component from DPDM. Table VII shows the overall performance, where DT and CL represent dynamic templates and contrastive learning, respectively. The symbol “-” denotes the removed component. As the training samples tend to be sufficient, both dynamic template and contrastive learning have positive effects on the performance. Meanwhile, with more training samples, contrastive learning plays a more important role than dynamic templates. When the training set tends to be small, the effect of the dynamic template is more remarkable. Further, the performances over various relations are visualized in Fig. 7. From this, it can be observed that DPDM is not superior to the baselines for all relations. Intuitively, through the utilization of dynamic templates and contrastive learning, the proposed DPDM model achieves a higher generalization ability at the slight expense of the performance of some well-performing relations. Such a situation indicates that the performance still has room for further improvement in the task of zero-shot relation extraction.

F. Alleviation for Representation Degradation (RQ-5)

The third preliminary study has mentioned the representation degradation problem, leading to relation representations that not guarantee alignment and uniformity. To confirm this problem is alleviated, we visualize the generated [MASK] embedding

of different relations in FewRel via *t-sne* technique. Both the visualizations without DPDM and with DPDM are shown in Fig. 8. Intuitively, [MASK] embedding without DPDM shows a conical distribution and varying concentration as mentioned in the preliminary study. In contrast, [MASK] embedding with DPDM exhibits a uniform distribution, and there are clearer boundaries among relations. Such a phenomenon suggests that the representation degradation problem has been alleviated by the proposed DPDM model.

VI. CONCLUSIONS AND FUTURE WORK

In this paper, we propose a novel dynamic Prompt-driven model for zero-shot relation extraction. Specifically, Prompt-based MLM can fully stimulate the knowledge from PLMs. Dynamic template in Prompt is leveraged to provide greater flexibility, and contrastive learning is introduced to further optimize the semantic representation. Extensive experiments on three benchmark datasets are conducted, demonstrating that our proposed model outperforms existing baselines significantly, even achieving better performance with fewer training samples.

In the future, we will further explore more effective Prompt-based strategies in the task of zero-shot relation extraction, to achieve improvement over all unseen relations, even some hard-to-extract relations with current low accuracy. In addition, since previous studies have demonstrated that introducing external data is effective for zero-shot learning as mentioned above, we also consider integrating external knowledge to realize data augmentation in the proposed model. The investigation of optimizing the Prompt paradigm to enhance model generalization in low-resource settings is a prominent issue that requires attention. Finally, the inner mechanism of Prompt, that is, how task-specific Prompt stimulates a pre-trained language model, is an interesting future work as well, which may theoretically promote the generalization ability of the proposed model.

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